

ORIGINAL ARTICLE

Targeting APOC3 with Olezarsen in Moderate Hypertriglyceridemia

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ABSTRACT

BACKGROUND

Highly effective therapies to reduce triglyceride levels are lacking. Olezarsen is an N-acetylgalactosamine–conjugated antisense oligonucleotide that targets the messenger RNA of apolipoprotein C-III, which inhibits triglyceride clearance.

METHODS

In this phase 3, international, double-blind, randomized, placebo-controlled trial, we enrolled patients with moderate hypertriglyceridemia (triglyceride level, 150 to 499 mg per deciliter) and elevated cardiovascular risk or with severe hypertriglyceridemia (triglyceride level, ≥ 500 mg per deciliter) and randomly assigned them in a 1:3 ratio to a 50-mg or 80-mg cohort. The patients were then randomly assigned in a 3:1 ratio to receive monthly subcutaneous olezarsen or matching placebo within each cohort. The primary outcome was the least-squares mean percent change in triglyceride level from baseline to 6 months among the patients with moderate hypertriglyceridemia, reported as the difference between each olezarsen dose group and the placebo group (the placebo-adjusted change).

RESULTS

A total of 1349 patients (254 in the olezarsen 50-mg group, 766 in the olezarsen 80-mg group, and 329 in the placebo group) were included in the primary efficacy analysis. The median age was 64 years, 40% of the patients were women, and the median triglyceride level at baseline was 238.5 mg per deciliter (interquartile range, 190.5 to 307.5). At 6 months, the placebo-adjusted least-squares mean change in triglyceride level was -58.4 percentage points (95% confidence interval [CI], -65.1 to -51.7 ; $P < 0.001$) in the olezarsen 50-mg group and -60.6 percentage points (95% CI, -67.1 to -54.0 ; $P < 0.001$) in the olezarsen 80-mg group. The incidence of serious adverse events appeared to be similar across the trial groups.

CONCLUSIONS

Among patients with moderate hypertriglyceridemia and elevated cardiovascular risk, treatment with olezarsen resulted in significantly greater reduction in triglyceride levels at 6 months than placebo. (Funded by Ionis Pharmaceuticals; ESSENCE–TIMI 73b ClinicalTrials.gov number, NCT05610280.)

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*A list of the Essence–TIMI 73b Investigators is provided in the Supplementary Appendix, available at NEJM.org.

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This article was published on August 30, 2025, at NEJM.org.

N Engl J Med 2025;393:1279–91.

DOI: 10.1056/NEJMoa2507227

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HYPERTRIGLYCERIDEMIA IS COMMON and is associated with an increased risk of atherosclerosis,¹⁻⁴ yet highly effective therapies for reducing triglyceride levels are lacking.⁵ Apolipoprotein C-III is a central regulator of triglyceride metabolism that limits the clearance of triglyceride-rich lipoproteins through inhibition of lipoprotein lipase and hepatic uptake of triglyceride-rich lipoprotein remnants.⁶⁻⁸ Olezarsen is an *N*-acetylgalactosamine–conjugated antisense oligonucleotide that targets *APOC3* messenger RNA.

Olezarsen has been shown to decrease triglyceride levels in small phase 2 trials^{9,10} and to reduce triglyceride levels and the risk of acute pancreatitis among patients with familial chylomicronemia syndrome.¹¹ The Essence–TIMI (Thrombolysis in Myocardial Infarction) 73b trial was designed to evaluate the efficacy and safety of olezarsen in patients with moderate hypertriglyceridemia who are at high cardiovascular risk and in those with severe hypertriglyceridemia.

METHODS

TRIAL DESIGN AND OVERSIGHT

In this phase 3, double-blind, randomized, placebo-controlled trial, we assessed olezarsen at a dose of 50 mg or 80 mg for lowering triglyceride levels. The trial design has been described previously,¹² and the protocol is available with the full text of this article at NEJM.org. The trial was designed in partnership between the TIMI Study Group and Ionis Pharmaceuticals, the trial sponsor. Data were captured at the participating-site level and held by a contract research organization (Medpace). The protocol was approved by all the relevant institutional ethical oversight committees. An independent data and safety monitoring board oversaw trial safety. The first draft of the manuscript was written by the first author, and all the authors made the decision to submit the manuscript for publication. The first author vouches for the completeness and accuracy of the data and for the fidelity of the trial to the protocol.

TRIAL POPULATION

Patients were eligible for the trial if they were at least 18 years of age and had either moderate hypertriglyceridemia (triglyceride level, 150 to 499 mg per deciliter [1.7 to 5.6 mmol per liter])

in addition to an elevated risk of cardiovascular events or severe hypertriglyceridemia (triglyceride level, ≥ 500 mg per deciliter [≥ 5.6 mmol per liter]). The minimum triglyceride level for eligibility was originally 150 mg per deciliter but was increased to 200 mg per deciliter (2.3 mmol per liter) with a protocol amendment that was made on June 27, 2023, on the basis of regulatory feedback that suggested the trial should be enriched for a population at increased cardiovascular risk.¹² At the time of enrollment, patients were expected to be receiving stable lipid-lowering therapy according to the local standard of care. Lifestyle interventions, including the limiting of alcohol intake, were encouraged before screening of potential patients.

Laboratory testing at the time of enrollment and during follow-up was performed after the patients had fasted. Eligible patients had to have a triglyceride level at both screening and enrollment of 150 mg per deciliter or higher (or ≥ 200 mg per deciliter after the protocol amendment). If the initial triglyceride level at screening or enrollment was below the minimum value, up to two additional tests were allowed, and the average of all the values obtained was used to assess eligibility.

The specific criteria for atherosclerotic cardiovascular disease or heightened cardiovascular risk have been described previously¹² and are provided in the Supplementary Appendix (available at NEJM.org). Preexisting atherosclerotic cardiovascular disease was defined as known coronary artery disease, cerebrovascular disease, or peripheral artery disease. Patients were considered to have an elevated risk of atherosclerotic cardiovascular disease if they had diabetes mellitus and were 55 years of age or older.

Among the key exclusion criteria were a recent acute coronary syndrome, cerebrovascular event, major surgery, or acute pancreatitis; relevant renal or hepatic laboratory abnormalities; or newly diagnosed or poorly controlled diabetes mellitus. Poorly controlled diabetes was defined by a glycated hemoglobin level of at least 9.5%.

RANDOMIZATION AND TREATMENT

Patients were randomly assigned in a 1:3 ratio to a 50-mg or 80-mg cohort and were then randomly assigned in a 3:1 ratio to receive subcutaneous olezarsen or volume-matched placebo within each cohort. Randomization was performed

with the use of commercial interactive response technology (Endpoint Clinical) and was stratified according to the triglyceride level used to determine eligibility (<500 or ≥ 500 mg per deciliter). The duration of the treatment period was 12 months and was followed by a 13-week safety follow-up period.

OUTCOMES

The primary outcome was the least-squares mean percent change in triglyceride level from baseline to 6 months among the patients with moderate hypertriglyceridemia, reported as the difference between each olezarsen dose group and the placebo group (the placebo-adjusted change). Secondary outcomes were the percent change in triglyceride level from baseline to 12 months, the proportion of patients who had a triglyceride level of less than 150 mg per deciliter at 6 and 12 months, and percent changes from baseline to 6 months and 12 months in the fasting levels of apolipoprotein C-III, very-low-density lipoprotein (VLDL) cholesterol, remnant cholesterol (the cholesterol on triglyceride-rich lipoproteins), non-high-density lipoprotein (HDL) cholesterol, HDL cholesterol, apolipoprotein B, and low-density lipoprotein (LDL) cholesterol. LDL and VLDL cholesterol were measured with ultracentrifugation, and remnant cholesterol was measured directly.¹³ All lipid levels at 6 months were based on the single laboratory specimen obtained at week 25. Major adverse cardiovascular events and acute pancreatitis (according to the Atlanta criteria¹⁴) were adjudicated centrally in a blinded manner. A complete list of the secondary and exploratory efficacy outcomes has been published previously¹² and is provided in the Supplementary Appendix. Assessments of safety outcomes included abnormalities in renal function, platelet counts, and liver-enzyme levels, as well as adverse clinical events. For patients assigned to the 80-mg cohort, a blinded dose reduction could be considered for safety concerns. Medpace Global Central Laboratories performed central testing of blood and urine specimens with the use of commercial assays.

STATISTICAL ANALYSIS

We determined that a sample size of 30 patients in each olezarsen group and 40 patients in the pooled placebo group would provide at least 90% power to show a 60-percentage-point difference

in the percent change in triglyceride levels from baseline to 6 months between each olezarsen dose group and the placebo group (the placebo-adjusted change), assuming a standard deviation of 73% and withdrawal of 20% of the patients, at a two-sided alpha level of 0.05. In addition to ensuring sufficient power for efficacy assessment, the number of patients enrolled also needed to account for regulatory requirements for total patient-years of exposure to olezarsen for safety assessments. We determined that a sample of 1475 patients, with three quarters assigned to the olezarsen groups, would fulfill both the efficacy and safety aims of the trial. No interim efficacy analysis was performed.

The primary outcome was evaluated according to the randomly assigned group among patients with a baseline triglyceride level of less than 500 mg per deciliter who received at least one dose of olezarsen or placebo (the primary efficacy population). We used an analysis of covariance model that included baseline triglyceride level. This primary efficacy population was defined in response to regulatory feedback that emphasized the importance of assessing the efficacy of olezarsen specifically in patients with moderate hypertriglyceridemia who were at high cardiovascular risk.¹² Patients with severe hypertriglyceridemia are the focus of the ongoing CORE-TIMI 72a and CORE2-TIMI 72b trials.¹⁵

For the assessment of lipid outcomes, we used a retrieved dropout-based multiple-imputation approach to account for missing lipid levels¹⁶; the approach is described in the statistical analysis plan (available with the protocol). We generated least-squares mean estimates of the percent change from baseline to provide an estimate of the difference between each olezarsen dose group and the placebo group, the 95% confidence interval, and the P value. The secondary outcomes were tested hierarchically in the primary efficacy population to control for multiple comparisons. If the result of the primary-outcome comparison was significant, then testing proceeded to the first outcome in the secondary-outcome hierarchy and continued sequentially until the first nonsignificant result was observed. The complete testing hierarchy is provided in the statistical analysis plan.

Continuous lipid values were analyzed with the use of an analysis-of-covariance model that included baseline lipid levels. Categorical triglyc-

eride level outcomes were analyzed with the use of a logistic model that included baseline triglyceride values. The safety population and the secondary efficacy population were defined as all the patients who received at least one dose of olezarsen or placebo. The secondary efficacy outcomes were tested in this secondary efficacy population in an exploratory fashion. We assessed safety events with a chi-square test; for events that were observed in fewer than five patients in

Table 1. Characteristics of the Patients at Baseline (Primary Efficacy Population).*

Characteristic	Olezarsen, 50 mg (N=254)	Olezarsen, 80 mg (N=766)	Placebo (N=329)
Median age (IQR) — yr	63 (57–69)	64 (56–70)	63 (56–70)
Female sex — no. (%)	97 (38.2)	306 (39.9)	140 (42.6)
Race and ethnic group — no. (%)†			
White	240 (94.5)	713 (93.1)	299 (90.9)
Black	2 (0.8)	34 (4.4)	16 (4.9)
Asian	6 (2.4)	8 (1.0)	6 (1.8)
Hispanic or Latino	67 (26.4)	179 (23.4)	68 (20.7)
Median body-mass index (IQR)‡	31.6 (28.3–34.9)	31.5 (28.5–35.9)	31.8 (28.1–35.2)
Atherosclerotic cardiovascular disease — no. (%)	115 (45.3)	307 (40.1)	128 (38.9)
Diabetes mellitus — no. (%)	151 (59.4)	468 (61.1)	190 (57.8)
Chronic kidney disease — no. (%)	27 (10.6)	64 (8.4)	30 (9.1)
Median triglyceride level (IQR) — mg/dl	235.3 (186.5–309.5)	237.3 (191.5–306.0)	244.0 (191.5–306.5)
Median apolipoprotein C-III level (IQR) — mg/dl	15.2 (12.3–18.8)	15.3 (12.9–18.3)	15.4 (12.7–18.8)
Median VLDL cholesterol level (IQR) — mg/dl	40.5 (30.7–53.0)	42.0 (32.7–53.0)	41.0 (31.5–55.0)
Median remnant cholesterol level (IQR) — mg/dl	50.7 (36.5–70.3)	49.4 (36.3–70.2)	48.8 (34.1–69.6)
Median non-HDL cholesterol level (IQR) — mg/dl	123.3 (97.5–158.0)	126.0 (99.5–155.5)	130.5 (106.0–167.0)
Median apolipoprotein B level (IQR) — mg/dl	89.5 (73.1–110.3)	90.7 (74.0–109.0)	94.5 (78.2–116.5)
Median LDL cholesterol level (IQR) — mg/dl	79.0 (58.0–112.0)	81.0 (60.0–107.3)	87.0 (66.0–117.5)
Median glycated hemoglobin level (IQR) — %	6.3 (5.7–7.1)	6.2 (5.7–7.0)	6.2 (5.6–7.1)
Median creatinine level (IQR) — mg/dl	0.9 (0.8–1.1)	0.9 (0.8–1.1)	0.9 (0.8–1.1)
Median platelet count (IQR) — $\times 10^3/\mu\text{l}$	236.3 (196.3–273.0)	232.8 (194.7–271.0)	232.0 (198.7–282.0)
Lipid-lowering therapy — no. (%)			
Any	243 (95.7)	739 (96.5)	317 (96.4)
Statin	200 (78.7)	622 (81.2)	265 (80.5)
Ezetimibe	43 (16.9)	127 (16.6)	59 (17.9)
Fibrate	55 (21.7)	172 (22.5)	82 (24.9)
N-3 fatty acid	65 (25.6)	172 (22.5)	64 (19.5)
Niacin	1 (0.4)	7 (0.9)	4 (1.2)
PCSK9 inhibitor	11 (4.3)	38 (5.0)	13 (4.0)
≥ 2 therapies	100 (39.4)	326 (42.6)	138 (41.9)

* The primary efficacy population included the patients with a baseline triglyceride level of less than 500 mg per deciliter who received at least one dose of olezarsen or placebo. Remnant cholesterol is the cholesterol on triglyceride-rich lipoproteins. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. To convert the values for creatinine to micromoles per liter, multiply by 88.4. HDL denotes high-density lipoprotein, IQR interquartile range, LDL low-density lipoprotein, PCSK9 proprotein convertase subtilisin-kexin type 9, and VLDL very-low-density lipoprotein.

† Race and ethnic group were reported by the patients. Patients could select more than one race or ethnic group.

‡ The body-mass index is the weight in kilograms divided by the square of the height in meters.

any group, a Fisher's exact test was used. There was no adjustment for competing risk. All the analyses were performed by the TIMI Study Group on a complete, independently held dataset with the use of SAS software, version 9.4 (SAS Institute).

RESULTS

PATIENTS

Patients were recruited at 160 sites in North America and Europe from November 2022 through

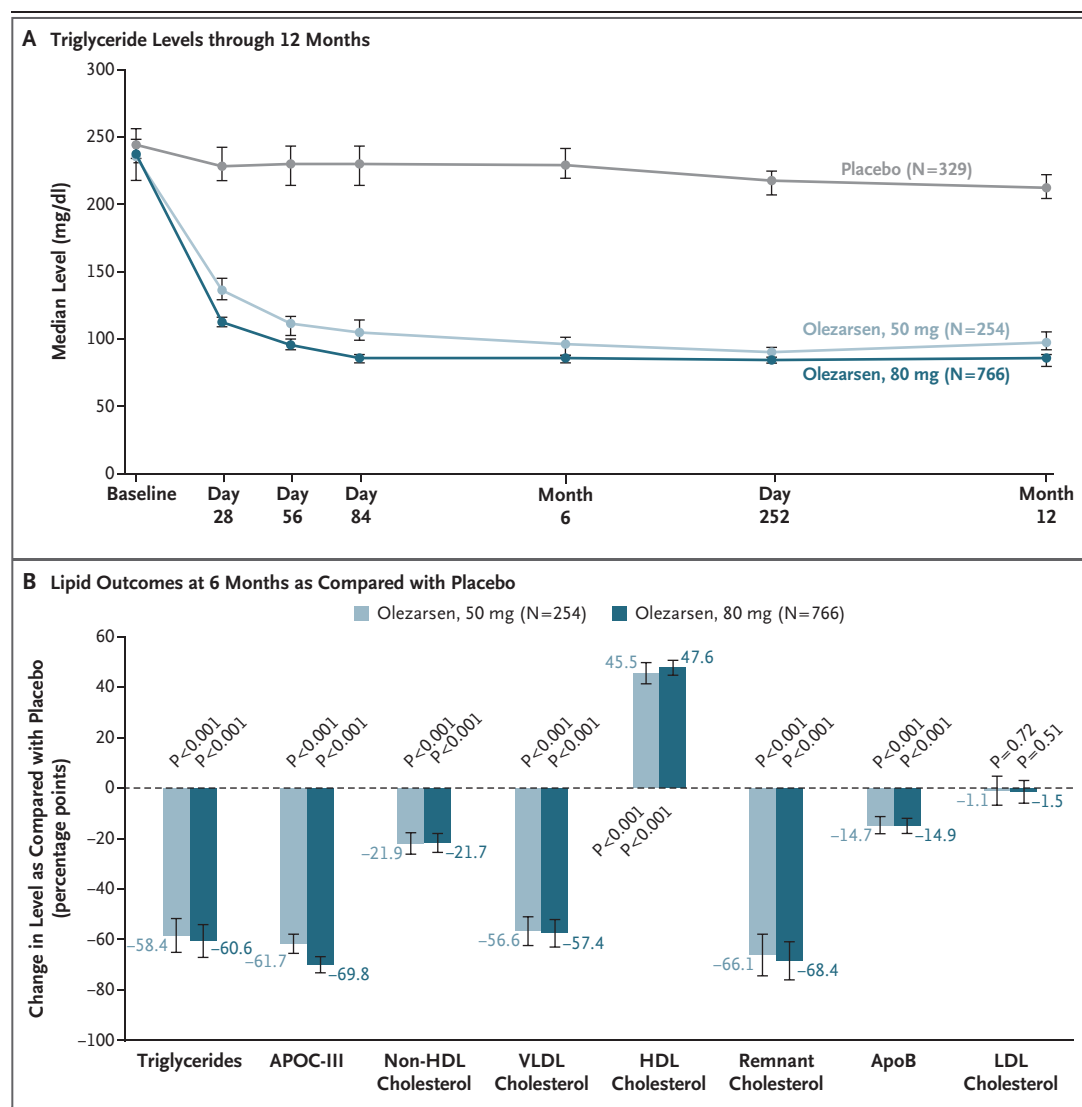


Figure 1. Triglyceride Levels through 12 Months and Lipid Outcomes at 6 Months.

Panel A shows the median triglyceride levels from baseline through 12 months. Panel B shows the primary outcome (the change in triglyceride levels) and secondary lipid outcomes at 6 months; the values are the least-squares mean differences in the percent change in levels from baseline between each olezarsen dose group and the placebo group. Data shown in both panels are for the primary efficacy population, which included the patients with a baseline triglyceride level of less than 500 mg per deciliter who received at least one dose of olezarsen or placebo. I bars indicate 95% confidence intervals. Remnant cholesterol is the cholesterol on triglyceride-rich lipoproteins. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. ApoB denotes apolipoprotein B, APOC-III apolipoprotein C-III, HDL high-density lipoprotein, LDL low-density lipoprotein, and VLDL very-low-density lipoprotein.

Table 2. Lipid Levels at Baseline and at 6 Months (Primary Efficacy Population).*

Variable	Olezarsen, 50 mg (N=254)	Olezarsen, 80 mg (N=766)	Placebo (N=329)
Triglycerides			
Level at baseline — mg/dl	258.7±86.9	257.4±85.5	260.0±84.9
Level at 6 mo — mg/dl	109.8±56.3	104.8±76.9	260.1±177.1
LSM change from baseline (95% CI) — %	−55.6 (−58.0 to −53.2)	−57.8 (−59.6 to −55.9)	2.8 (−3.5 to 9.0)
Placebo-adjusted LSM change (95% CI) — percentage points	−58.4 (−65.1 to −51.7)	−60.6 (−67.1 to −54.0)	—
P value vs. placebo	<0.001	<0.001	—
Apolipoprotein C-III			
Level at baseline — mg/dl	16.1±5.1	16.2±4.8	16.2±4.6
Level at 6 mo — mg/dl	6.0±3.2	4.8±3.6	16.0±5.7
LSM change from baseline (95% CI) — %	−61.7 (−64.2 to −59.2)	−69.8 (−71.4 to −68.2)	0.0 (−2.8 to 2.8)
Placebo-adjusted LSM change (95% CI) — percentage points	−61.7 (−65.5 to −57.9)	−69.8 (−73.0 to −66.6)	—
P value vs. placebo	<0.001	<0.001	—
Non-HDL cholesterol			
Level at baseline — mg/dl	130.6±43.5	132.2±44.8	139.3±46.1
Level at 6 mo — mg/dl	102.6±40.2	104.0±44.2	138.3±51.4
LSM change from baseline (95% CI) — %	−19.9 (−22.6 to −17.2)	−19.7 (−21.7 to −17.7)	2.0 (−1.2 to 5.2)
Placebo-adjusted LSM change (95% CI) — percentage points	−21.9 (−26.1 to −17.7)	−21.7 (−25.5 to −17.9)	—
P value vs. placebo	<0.001	<0.001	—
VLDL cholesterol			
Level at baseline — mg/dl	44.0±16.4	44.9±17.7	45.7±19.4
Level at 6 mo — mg/dl	21.4±10.7	21.2±14.2	46.8±25.8
LSM change from baseline (95% CI) — %	−48.7 (−51.4 to −46.1)	−49.5 (−51.7 to −47.4)	7.9 (2.8 to 12.9)
Placebo-adjusted LSM change (95% CI) — percentage points	−56.6 (−62.3 to −50.9)	−57.4 (−62.9 to −52.0)	—
P value vs. placebo	<0.001	<0.001	—
HDL cholesterol			
Level at baseline — mg/dl	39.6±11.7	39.7±10.8	39.7±10.0
Level at 6 mo — mg/dl	58.7±18.6	59.9±18.5	41.2±11.0
LSM change from baseline (95% CI) — %	50.4 (46.6 to 54.1)	52.5 (50.3 to 54.7)	4.9 (2.9 to 6.9)
Placebo-adjusted LSM change (95% CI) — percentage points	45.5 (41.2 to 49.7)	47.6 (44.7 to 50.5)	—
P value vs. placebo	<0.001	<0.001	—
Remnant cholesterol			
Level at baseline — mg/dl	56.7±28.8	57.4±31.5	57.8±34.6
Level at 6 mo — mg/dl	21.6±13.3	21.8±20.5	56.0±36.2
LSM change from baseline (95% CI) — %	−55.5 (−59.3 to −51.7)	−57.7 (−59.9 to −55.5)	10.7 (3.4 to 17.9)
Placebo-adjusted LSM change (95% CI) — percentage points	−66.1 (−74.3 to −57.9)	−68.4 (−76.0 to −60.8)	—
P value vs. placebo	<0.001	<0.001	—

Table 2. (Continued.)

Variable	Olezarsen, 50 mg (N=254)	Olezarsen, 80 mg (N=766)	Placebo (N=329)
Apolipoprotein B			
Level at baseline — mg/dl	93.2±27.0	93.9±27.7	98.7±27.7
Level at 6 mo — mg/dl	78.3±24.1	78.8±26.9	97.1±30.3
LSM change from baseline (95% CI) — %	−14.4 (−16.8 to −12.0)	−14.6 (−16.3 to −13.0)	0.3 (−2.1 to 2.8)
Placebo-adjusted LSM change (95% CI) — percentage points	−14.7 (−18.1 to −11.3)	−14.9 (−17.9 to −11.9)	—
P value vs. placebo	<0.001	<0.001	—
LDL cholesterol			
Level at baseline — mg/dl	86.6±37.7	87.3±38.3	93.9±38.5
Level at 6 mo — mg/dl	82.8±39.0	83.2±39.0	91.4±41.1
LSM change from baseline (95% CI) — %	0.1 (−4.5 to 4.7)	−0.4 (−3.2 to 2.4)	1.1 (−2.4 to 4.7)
Placebo-adjusted LSM change (95% CI) — percentage points	−1.1 (−6.9 to 4.7)	−1.5 (−6.0 to 3.0)	—
P value vs. placebo	0.72	0.51	—

* Plus-minus values are means ±SD. The least-squares mean (LSM) changes from baseline are the changes from baseline to 6 months. The placebo-adjusted LSM change is the LSM difference in the percent change from baseline to 6 months between each olezarsen dose group and the placebo group.

February 2024, and 1478 patients underwent randomization. A total of 1349 patients (91.3%) had triglyceride levels of less than 500 mg per deciliter at baseline and received at least one dose of olezarsen or placebo; therefore, the primary efficacy population included 254 patients in the olezarsen 50-mg group, 766 in the olezarsen 80-mg group, and 329 in the placebo group (Fig. S1 in the Supplementary Appendix). Early discontinuation of olezarsen or placebo occurred in 9.7% of the patients in the olezarsen 50-mg group, 14.8% of those in the olezarsen 80-mg group, and 11.4% of those in the placebo group. A total of 8 patients (0.5%) were lost to follow-up, 10 (0.7%) withdrew consent, and 9 (0.6%) died during the trial.

The baseline characteristics of the patients in the primary efficacy population are provided in Table 1. The median age was 64 years (interquartile range, 56 to 70), 40.3% were women, 92.8% were White, 3.9% were Black, and 23.3% identified as Hispanic or Latino. A total of 60.0% of the patients had diabetes mellitus, and 96.3% were receiving at least one lipid-lowering therapy. The median triglyceride level at baseline was 238.5 mg per deciliter (2.7 mmol per liter; interquartile range, 190.5 to 307.5 mg per deciliter [2.2 to 3.5 mmol per liter]). The characteristics of the patients

at baseline appeared to be balanced across the groups, although levels of non-HDL cholesterol, apolipoprotein B, and LDL cholesterol were slightly higher among the patients in the placebo group than among those in the olezarsen groups. The representativeness of the trial population relative to the broader population with hypertriglyceridemia is shown in Table S1. The baseline characteristics of the patients in the safety population are provided in Table S2.

OUTCOMES

At 6 months, the least-squares mean change from baseline in triglyceride level was −55.6% (95% confidence interval [CI], −58.0 to −53.2) in the olezarsen 50-mg group, −57.8% (95% CI, −59.6 to −55.9) in the olezarsen 80-mg group, and 2.8% (95% CI, −3.5 to 9.0) in the placebo group. The placebo-adjusted least-squares mean change in triglyceride level from baseline to 6 months was −58.4 percentage points (95% CI, −65.1 to −51.7; $P<0.001$) in the olezarsen 50-mg group and −60.6 percentage points (95% CI, −67.1 to −54.0; $P<0.001$) in the olezarsen 80-mg group (Fig. 1 and Table 2). The effects of olezarsen on triglyceride levels in key clinical subgroups are shown in Figure 2.

At 12 months, the placebo-adjusted least-

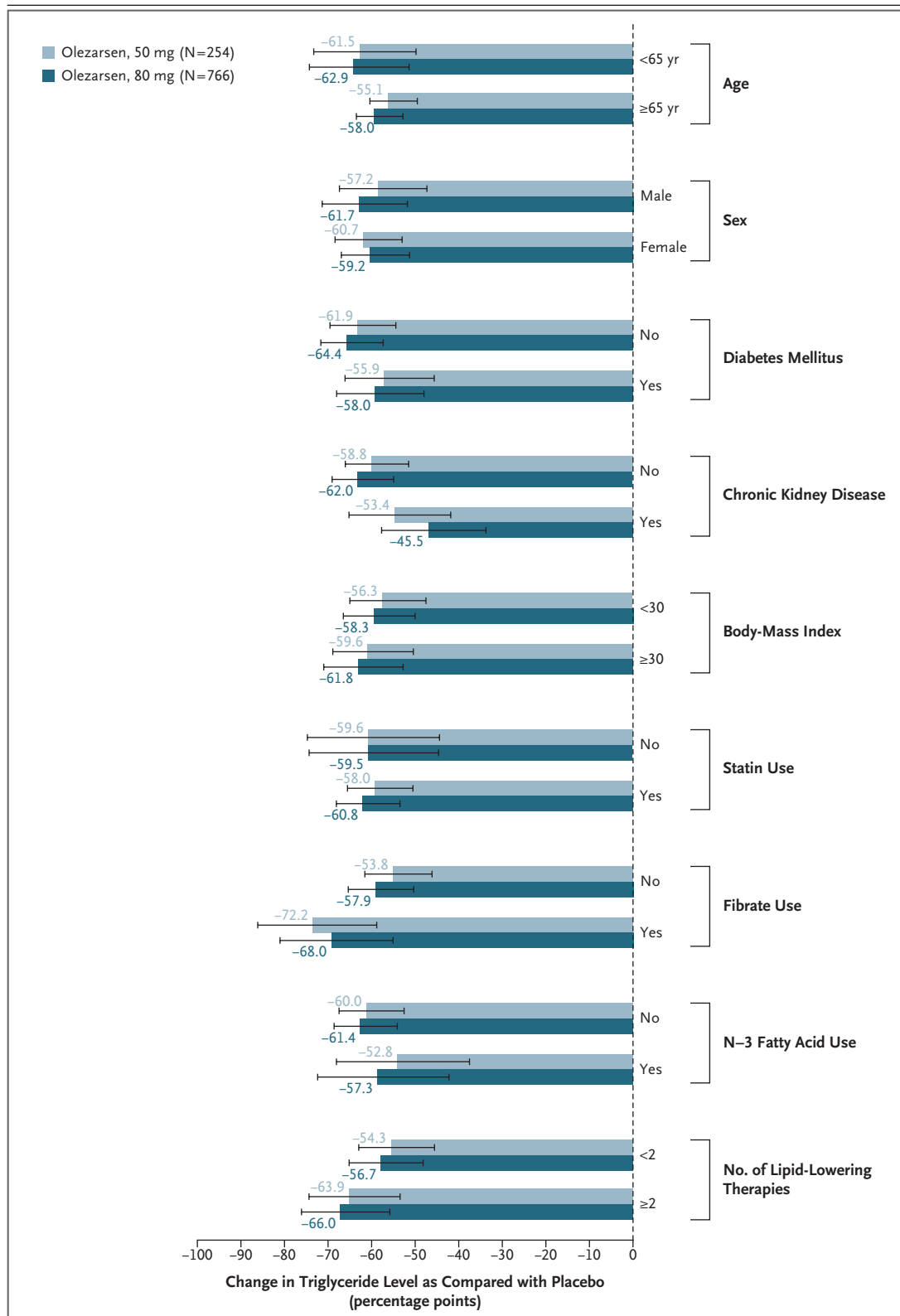


Figure 2 (facing page). Changes in Triglyceride Levels at 6 Months in Key Clinical Subgroups.

Shown are the least-squares mean differences in the percent change in triglyceride levels from baseline to 6 months between each olezarsen dose group and the placebo group. The body-mass index is the weight in kilograms divided by the square of the height in meters. I bars indicate 95% confidence intervals.

squares mean change in triglyceride level was -50.7 percentage points (95% CI, -60.6 to -40.7 ; $P < 0.001$) in the olezarsen 50-mg group and -50.7 percentage points (95% CI, -60.3 to -41.1 ; $P < 0.001$) in the olezarsen 80-mg group (Table S3). At 6 months, 85.0% of the patients in the olezarsen 50-mg group, 88.7% of those in the olezarsen 80-mg group, and 12.5% of those in the placebo group had a triglyceride level of less than 150 mg per deciliter ($P < 0.001$ for both olezarsen groups vs. placebo) (Fig. S2). At 12 months, the percentage of patients who had a fasting triglyceride level of less than 150 mg per deciliter was 82.8% in the olezarsen 50-mg group, 85.0% in the olezarsen 80-mg group, and 20.6% in the placebo group ($P < 0.001$ for each olezarsen group vs. placebo).

The changes in levels of apolipoprotein C-III, non-HDL cholesterol, HDL cholesterol, VLDL cholesterol, directly measured remnant cholesterol, apolipoprotein B, and LDL cholesterol at 6 and 12 months are shown in Table 2 and Table S3. There was no apparent change from baseline in LDL cholesterol level as compared with placebo with either dose of olezarsen. The findings for the primary and secondary outcomes in the secondary efficacy population appeared to be consistent with those in the primary efficacy population (Table S4); the findings in analyses that included only patients with measured lipid levels at 6 months were also consistent with those in the primary efficacy population (Table S5). In the primary efficacy population, a major adverse cardiovascular event occurred in 7 patients (2.8%) in the olezarsen 50-mg group, 44 (5.7%) in the olezarsen 80-mg group, and 14 (4.3%) in the placebo group through 12 months (Table S6). Acute pancreatitis occurred in no patients in the olezarsen 50-mg group, in 3 patients (0.4%) in the olezarsen 80-mg group, and in no patients in the placebo group.

SAFETY

The frequency of adverse events during the treatment period and through 28 days after the last dose of olezarsen or placebo was similar across the trial groups (Table 3). The frequency of serious adverse events appeared to be balanced across the trial groups, as were serious adverse events resulting in discontinuation of olezarsen or placebo. Injection-site reactions were more common with olezarsen than with placebo and were generally mild in severity (Table 3).

There were no significant differences among the groups with respect to elevations in alanine aminotransferase or aspartate aminotransferase levels to at least three times or five times the upper limit of the normal range (Table 3). Mild elevations in aminotransferase levels to any degree above the upper limit of the normal range were more common in the olezarsen 50-mg group (34.2%) and the olezarsen 80-mg group (38.3%) than in the placebo group (17.6%) ($P < 0.001$ for both olezarsen groups vs. placebo). However, there were no significant between-group differences in elevations in aminotransferase levels to more than 3 or 5 times the upper limit of the normal range. No case met Hy's law criteria (which would indicate a high risk of severe drug-induced liver injury).

A platelet count of less than 100,000 per microliter occurred in 0.8% of the patients in the placebo group, in 2.2% of those in the olezarsen 50-mg group ($P = 0.18$ vs. placebo), and in 2.3% of those in the olezarsen 80-mg group ($P = 0.10$ vs. placebo). A platelet count of less than 75,000 per microliter occurred in 1% or less of the patients in each group (Table 3), and no patients discontinued olezarsen or placebo according to the thrombocytopenia stopping rules (details are provided in the Supplementary Appendix). Glycated hemoglobin levels at 6 months are shown in Table S7; an apparent increase in glycated hemoglobin levels was observed among patients with a diagnosis of diabetes mellitus at baseline. No apparent change was noted in insulin resistance (according to the homeostatic model assessment for insulin resistance) or beta-cell function with olezarsen as compared with placebo in patients with or without diabetes (Table S8).

Table 3. Key Safety and Laboratory Measures (Safety Population).*

Safety Measure	Placebo (N=369)	Olezarsen, 50 mg (N=276)	P Value vs. Placebo	Olezarsen, 80 mg (N=832)	P Value vs. Placebo
Adverse events — no. (%)					
Any adverse event	266 (72.1)	201 (72.8)	0.84	638 (76.7)	0.09
Adverse event leading to discontinuation of olezarsen or placebo	18 (4.9)	12 (4.3)	0.75	56 (6.7)	0.22
Any serious adverse event	42 (11.4)	26 (9.4)	0.42	113 (13.6)	0.29
Serious adverse event leading to discontinuation of olezarsen or placebo	5 (1.4)	2 (0.7)	0.70	18 (2.2)	0.49
Injection-site reaction†					
Any	6 (1.6)	42 (15.2)	<0.001	135 (16.2)	<0.001
Mild	6 (1.6)	37 (13.4)	<0.001	125 (15.0)	<0.001
Moderate	1 (0.3)	8 (2.9)	0.006	24 (2.9)	0.002
Severe	0	0	—	0	—
Possible hypersensitivity reaction‡	20 (5.4)	13 (4.7)	0.69	38 (4.6)	0.53
Laboratory measures — no./total no. (%)					
Hepatic abnormality					
ALT or AST level $\geq 3 \times$ ULN	3/363 (0.8)	7/272 (2.6)	0.11	17/822 (2.1)	0.15
ALT or AST level $\geq 5 \times$ ULN	1/363 (0.3)	2/272 (0.7)	0.58	2/822 (0.2)	0.99
Total bilirubin level $\geq 2 \times$ ULN	1/363 (0.3)	1/272 (0.4)	0.99	1/822 (0.1)	0.52
Renal abnormality					
Decrease in eGFR $\geq 30\%$	14/363 (3.9)	9/272 (3.3)	0.71	33/822 (4.0)	0.90
Decrease in eGFR $\geq 50\%$	2/363 (0.6)	1/272 (0.4)	0.99	7/822 (0.9)	0.73
Urinary protein:creatinine ratio $\geq 1000\text{§}$	7/363 (1.9)	4/272 (1.5)	0.77	16/822 (1.9)	0.98
Urinary protein:creatinine ratio $\geq 3000\text{§}$	0/363 (0)	1/272 (0.4)	0.43	1/822 (0.1)	0.99
Platelet count					
<100,000/ μl	3/363 (0.8)	6/272 (2.2)	0.18	19/821 (2.3)	0.10
<75,000/ μl	1/363 (0.3)	3/272 (1.1)	0.32	6/821 (0.7)	0.68
<50,000/ μl	1/363 (0.3)	0/272 (0)	0.99	1/821 (0.1)	0.52

* The safety population included all the patients who had received at least one dose of olezarsen or placebo. Data are shown through 28 days after the last dose of olezarsen or placebo. ALT denotes alanine aminotransferase, AST aspartate aminotransferase, eGFR estimated glomerular filtration rate, and ULN upper limit of the normal range.

† Patients could have had events in more than one category.

‡ Possible hypersensitivity reactions were defined according to *Medical Dictionary for Regulatory Activities*, version 28.0, standardized queries with the use of narrow terms.

§ Urinary protein was measured in milligrams, and urinary creatinine in grams.

DISCUSSION

In this phase 3 trial, olezarsen, as compared with placebo, significantly reduced levels of triglycerides, remnant cholesterol, non-HDL cholesterol, and apolipoprotein B without affecting LDL cholesterol levels among patients with moderate hypertriglyceridemia and elevated cardiovascular

risk who were receiving standard lipid-lowering therapy.

Hypertriglyceridemia is frequently identified in clinical practice, yet effective therapies are lacking.^{5,17,18} Moderate elevations in triglyceride levels are associated with an increased risk of atherosclerotic cardiovascular disease,^{1,2,19,20} and triglyceride-rich lipoproteins are thought to be at

least as atherogenic as LDL cholesterol on a per-particle basis.^{19,21} Furthermore, hypertriglyceridemia, when severe, can result in immediate clinical consequences, such as acute pancreatitis.²²

Existing pharmacologic therapies such as statins, prescription-strength n-3 fatty acids, fibrates, and ezetimibe have only modest effects on triglyceride levels, with typical reductions ranging from less than 10% up to approximately 30%.¹⁷ Olezarsen has been approved in the United States as an adjunct to diet to reduce triglyceride levels in adults with the rare familial chylomicronemia syndrome,¹¹ but its safety and efficacy in broader populations with elevated triglyceride levels are not fully known. In the current trial, we found that olezarsen resulted in substantially greater reductions in triglyceride levels than are typical with therapies currently available for this population, and nearly 90% of the patients with moderate hypertriglyceridemia who received olezarsen had normal triglyceride levels at 6 months and 12 months. In addition, we observed that, as compared with placebo, treatment with olezarsen resulted in significant reductions in remnant cholesterol levels (up to nearly 70%) as well as in the total number of atherogenic lipoproteins as assessed on the basis of the apolipoprotein B level, with no change in LDL cholesterol level. Determining whether these changes to the lipid profile would translate into a clinical benefit with respect to cardiovascular outcomes would require a dedicated trial.

Genetic studies indicate that patients with loss-of-function variants in *APOC3* have lower triglyceride levels as well as a lower risk of atherosclerosis than those who do not have such variants,⁶⁻⁸ and therefore inhibition of apolipoprotein C-III translation has emerged as an attractive strategy.^{9,10,23} The placebo-adjusted reductions in apolipoprotein C-III levels (62 to 70%) and in triglyceride levels (58 to 61%) observed in this trial appear to be similar to those seen with plzasiran, a small interfering RNA that targets apolipoprotein C-III, in a cohort of patients with mixed hyperlipidemia, which have ranged from approximately 57 to 79% for apolipoprotein C-III and approximately 44 to 62% for triglycerides.²³ The apparent potential interaction between fibrate use and reduction in triglyceride levels with olezarsen that we observed in this trial may have been attributable to chance rather than indicat-

ing heightened efficacy among patients who were receiving fibrate therapy; however, ongoing trials of olezarsen involving patients with severe hypertriglyceridemia may provide additional insight.¹⁵

Other targets for triglyceride lowering are being explored.^{3,18} Zodasiran, a small interfering RNA against angiopoietin-like protein 3, reduced triglyceride levels to a similar degree as olezarsen,²⁴ and reductions of 40 to 50% in triglyceride levels were seen with a monoclonal antibody against angiopoietin-like protein 4 in patients with moderate hypertriglyceridemia.²⁵ Pegzofermin, a fibroblast growth factor 21 analogue, reduced triglycerides by 29 to 53% in patients with severe hypertriglyceridemia.²⁶ In addition, agents intended for weight loss, such as glucagon-like peptide-1 receptor agonists, reduce triglyceride levels, although to a lesser degree than targeted therapies.²⁷ Triglyceride lowering is a rapidly evolving field with several therapeutic outcomes of potential interest, including effects on lipid levels, acute pancreatitis, hepatic steatosis, and cardiovascular events.

With respect to the safety profile of olezarsen, clinically meaningful increases in liver aminotransferase levels were rare and were similar in frequency across the trial groups. Studies of the parent compound of olezarsen, volanesorsen, suggest a potential reduction in hepatic steatosis among patients with severe hypertriglyceridemia²⁸; this potential reduction in hepatic fat content is being evaluated in the CORE-TIMI 72a and CORE2-TIMI 72b trials.¹⁵ There were small increases in glycated hemoglobin levels among patients with preexisting diabetes. Mild elevations in aminotransferase levels as well as small increases in glycated hemoglobin levels have also been seen with plzasiran^{18,23,29} and are similar to the now well-established safety profile of statins.^{30,31}

Our trial has limitations. First, the trial was primarily designed to assess olezarsen treatment among patients with moderate hypertriglyceridemia rather than across the full range of triglyceride levels. Two phase 3 trials¹⁵ are specifically evaluating olezarsen among patients with severe hypertriglyceridemia. Second, the efficacy and safety of olezarsen beyond 1 year of treatment were not assessed. A longer-term open-label extension program among patients with severe hypertriglyceridemia is under way.¹⁵ Third, the

majority of the patients in this trial were White, and results may not be generalizable to other races and ethnic groups.

Among patients with moderate hypertriglyceridemia and elevated cardiovascular risk, monthly treatment with olezarsen reduced triglyceride levels at 6 months as compared with placebo.

Supported by Ionis Pharmaceuticals.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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REFERENCES

- Marston NA, Giugliano RP, Im K, et al. Association between triglyceride lowering and reduction of cardiovascular risk across multiple lipid-lowering therapeutic classes: a systematic review and meta-regression analysis of randomized controlled trials. *Circulation* 2019;140:1308-17.
- Nordestgaard BG, Varbo A. Triglycerides and cardiovascular disease. *Lancet* 2014;384:626-35.
- Bergmark BA, Marston NA, Bramson CR, et al. Effect of vupanorsen on non-high-density lipoprotein cholesterol levels in statin-treated patients with elevated cholesterol: TRANSLATE-TIMI 70. *Circulation* 2022;145:1377-86.
- Nordestgaard AT, Pradhan AD, Everett BM, et al. Expanding the triglyceride range in clinical trials: therapeutic opportunities. *Eur Heart J* 2025;46:1835-48.
- Ginsberg HN, Packard CJ, Chapman MJ, et al. Triglyceride-rich lipoproteins and their remnants: metabolic insights, role in atherosclerotic cardiovascular disease, and emerging therapeutic strategies — a consensus statement from the European Atherosclerosis Society. *Eur Heart J* 2021;42:4791-806.
- Jørgensen AB, Frikke-Schmidt R, Nordestgaard BG, Tybjaerg-Hansen A. Loss-of-function mutations in *APOC3* and risk of ischemic vascular disease. *N Engl J Med* 2014;371:32-41.
- Pollin TI, Damcott CM, Shen H, et al. A null mutation in human *APOC3* confers a favorable plasma lipid profile and apparent cardioprotection. *Science* 2008;322:1702-5.
- The TG and HDL Working Group of the Exome Sequencing Project, National Heart, Lung, and Blood Institute. Loss-of-function mutations in *APOC3*, triglycerides, and coronary disease. *N Engl J Med* 2014;371:22-31.
- Bergmark BA, Marston NA, Prohaska TA, et al. Olezarsen for hypertriglyceridemia in patients at high cardiovascular risk. *N Engl J Med* 2024;390:1770-80.
- Tardif J-C, Karwatowska-Prokopczuk E, St Amour E, et al. Apolipoprotein C-III reduction in subjects with moderate hypertriglyceridaemia and at high cardiovascular risk. *Eur Heart J* 2022;43:1401-12.
- Stroes ESG, Alexander VJ, Karwatowska-Prokopczuk E, et al. Olezarsen, acute pancreatitis, and familial chylomicronemia syndrome. *N Engl J Med* 2024;390:1781-92.
- Bergmark BA, Marston NA, Prohaska TA, et al. Olezarsen in patients with hypertriglyceridemia at high cardiovascular risk: rationale and design of the Essence-TIMI 73b trial. *Am Heart J* 2025;286:116-24.
- Varbo A, Nordestgaard BG. Directly measured vs. calculated remnant cholesterol identifies additional overlooked individuals in the general population at higher risk of myocardial infarction. *Eur Heart J* 2021;42:4833-43.
- Banks PA, Bollen TL, Derveniz C, et al. Classification of acute pancreatitis — 2012: revision of the Atlanta classification and definitions by international consensus. *Gut* 2013;62:102-11.
- Marston NA, Bergmark BA, Alexander VJ, et al. Design and rationale of the CORE-TIMI 72a and CORE2-TIMI 72b trials of olezarsen in patients with severe hypertriglyceridemia. *Am Heart J* 2025;286:125-35.
- Carpenter JR, Roger JH, Kenward MG. Analysis of longitudinal trials with protocol deviation: a framework for relevant, accessible assumptions, and inference via multiple imputation. *J Biopharm Stat* 2013;23:1352-71.
- Mach F, Baigent C, Catapano AL, et al. 2019 ESC/EAS guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *Eur Heart J* 2020;41:111-88.
- Nordestgaard AT, Tybjaerg-Hansen A, Mansbach H, Kersten S, Nordestgaard BG, Rosenson RS. Target populations for novel triglyceride-lowering therapies. *J Am Coll Cardiol* 2025;85:1876-97.
- Björnsön E, Adiels M, Taskinen M-R, et al. Triglyceride-rich lipoprotein remnants, low-density lipoproteins, and risk of coronary heart disease: a UK Biobank study. *Eur Heart J* 2023;44:4186-95.
- Ference BA, Kastelein JJP, Ray KK, et al. Association of triglyceride-lowering LPL variants and LDL-C-lowering LDLR variants with risk of coronary heart disease. *JAMA* 2019;321:364-73.
- Nordestgaard BG. Triglyceride-rich lipoproteins and atherosclerotic cardiovascular disease: new insights from epidemiology, genetics, and biology. *Circ Res* 2016;118:547-63.
- Hansen SEJ, Varbo A, Nordestgaard BG, Langsted A. Hypertriglyceridemia-associated pancreatitis: new concepts and potential mechanisms. *Clin Chem* 2023;69:1132-44.
- Ballantyne CM, Vasas S, Azizad M, et al. Plozasiran, an RNA interference agent targeting APOC3, for mixed hyperlipidemia. *N Engl J Med* 2024;391:899-912.
- Rosenson RS, Gaudet D, Hegele RA, et al. Zodasiran, an RNAi therapeutic targeting ANGPTL3, for mixed hyperlipidemia. *N Engl J Med* 2024;391:913-25.
- Cummings BB, Joing MP, Bouchard PR, et al. Safety and efficacy of a novel ANGPTL4 inhibitory antibody for lipid lowering: results from phase 1 and phase 1b/2a clinical studies. *Lancet* 2025;405:1923-34.
- Bhatt DL, Bays HE, Miller M, et al. The FGF21 analog pegozafermin in severe hypertriglyceridemia: a randomized phase 2 trial. *Nat Med* 2023;29:1782-92.
- Linetzky B, Sattar N, Verma S, et al. Improvements in cardiometabolic risk factors by weight reduction: a post hoc analysis of adults with obesity randomly assigned to tirzepatide. *Ann Intern Med* 2025 June 24 (Epub ahead of print).

28. Prohaska TA, Alexander VJ, Karwowska-Prokopczuk E, et al. APOC3 inhibition with volanesorsen reduces hepatic steatosis in patients with severe hypertriglyceridemia. *J Clin Lipidol* 2023;17:406-11.
29. Gaudet D, Pall D, Watts GF, et al. Plozasiran (ARO-APOC3) for severe hypertriglyceridemia: the SHASTA-2 randomized clinical trial. *JAMA Cardiol* 2024;9:620-30.
30. Adhyaru BB, Jacobson TA. Safety and efficacy of statin therapy. *Nat Rev Cardiol* 2018;15:757-69.
31. Reith C, Preiss D, Blackwell L, et al. Effects of statin therapy on diagnoses of new-onset diabetes and worsening glycaemia in large-scale randomised blinded statin trials: an individual participant data meta-analysis. *Lancet Diabetes Endocrinol* 2024;12:306-19.

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